WORKED SOLUTION

STEP III Specimen paper, Question 1

Pure • Proof by Induction

STEP III Specimen paper, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q1

The key idea

For (i), work out the first few products and look at them the right way: once each one is written with denominator $2n$, the pattern jumps out. Factorising $1-\frac{1}{m^2}=\frac{(m-1)(m+1)}{m^2}$ explains why, and makes the inductive step a one-line calculation. For (ii), induct on k with n fixed: adding one more term to the sum is exactly Pascal's rule in disguise.

Solution

Part (i)

Write $P_n=\left(1-\frac{1}{4}\right)\left(1-\frac{1}{9}\right)\cdots\left(1-\frac{1}{n^2}\right)$. The first few values are

$$P_2=\frac{3}{4},\qquad P_3=\frac{3}{4}\cdot\frac{8}{9}=\frac{2}{3},\qquad P_4=\frac{2}{3}\cdot\frac{15}{16}=\frac{5}{8},\qquad P_5=\frac{5}{8}\cdot\frac{24}{25}=\frac{3}{5}.$$

The denominators 4, 8 suggest writing everything over $2n$: $P_2=\frac{3}{4}$, $P_3=\frac{4}{6}$, $P_4=\frac{5}{8}$, $P_5=\frac{6}{10}$. So we guess

$$P_n=\frac{n+1}{2n}\qquad\text{for } n\geq 2.$$

Proof by induction. Let $S(n)$ be the statement $P_n=\dfrac{n+1}{2n}$.

Base case. $P_2=1-\frac{1}{4}=\frac{3}{4}=\frac{2+1}{2\cdot 2}$, so $S(2)$ is true.

Inductive step. Suppose $S(n)$ is true for some $n\geq 2$. Then, since $P_{n+1}=P_n\left(1-\frac{1}{(n+1)^2}\right)$ and $1-\frac{1}{(n+1)^2}=\frac{(n+1)^2-1}{(n+1)^2}=\frac{n(n+2)}{(n+1)^2}$,

$$\begin{aligned}P_{n+1}&=\frac{n+1}{2n}\cdot\frac{n(n+2)}{(n+1)^2}\\&=\frac{n+2}{2(n+1)},\end{aligned}$$

which is $S(n+1)$. By induction, $S(n)$ holds for all $n \geq 2$.

(The factorisation also explains the guess: $P_n = \prod_{m=2}^{n} \frac{m-1}{m} \cdot \prod_{m=2}^{n} \frac{m+1}{m} = \frac{1}{n} \cdot \frac{n+1}{2}$, both products telescoping.)

Part (ii)

Fix a positive integer n, and let $T(k)$ be the statement

$$\sum_{r=0}^{k} (-1)^r \binom{n}{r} = (-1)^k \binom{n-1}{k}.$$

We prove $T(k)$ for $k = 0, 1, \ldots, n-1$ by induction on k.

Base case. For $k = 0$ both sides equal $\binom{n}{0} = \binom{n-1}{0} = 1$.

Inductive step. Suppose $T(k)$ holds for some k with $0 \leq k \leq n-2$ (so that $k+1$ is still in the range). Adding the next term,

$$\begin{aligned} \sum_{r=0}^{k+1} (-1)^r \binom{n}{r} &= (-1)^k \binom{n-1}{k} + (-1)^{k+1} \binom{n}{k+1} \\ &= (-1)^{k+1} \left[\binom{n}{k+1} - \binom{n-1}{k} \right]. \end{aligned}$$

? **Ask yourself** What is $\binom{n}{k+1} - \binom{n-1}{k}$?

Pascal's rule says $\binom{n}{k+1} = \binom{n-1}{k} + \binom{n-1}{k+1}$: choosing $k+1$ objects from n, either a particular object is chosen (and k more come from the other $n-1$) or it isn't (and all $k+1$ come from the other $n-1$). So the bracket is $\binom{n-1}{k+1}$, and

$$\sum_{r=0}^{k+1} (-1)^r \binom{n}{r} = (-1)^{k+1} \binom{n-1}{k+1},$$

which is $T(k+1)$. By induction, $T(k)$ holds for $k = 0, 1, \ldots, n-1$, as required.

If $n = 1$ the only case is $k = 0$, covered by the base case, so the argument needs no special treatment there.

Checks

- $P_{300} = \frac{301}{600} \approx 0.5017$, and a computer multiplying the factors exactly with fractions agreed with $\frac{n+1}{2n}$ for every n from 2 to 300. As $n \to \infty$ the product tends to $\frac{1}{2}$.
- For $n = 4$: the partial sums $1, 1-4 = -3, -3+6 = 3, 3-4 = -1$ match $\binom{3}{0}, -\binom{3}{1}, \binom{3}{2}, -\binom{3}{3}$. The computer checked (ii) exactly for all $n \leq 79$ and all $k \leq n-1$.
- Taking $k = n-1$ and adding the last term $(-1)^n \binom{n}{n}$ gives $(-1)^{n-1} + (-1)^n = 0$, which is the familiar fact $\sum_{r=0}^{n} (-1)^r \binom{n}{r} = (1-1)^n = 0$.

Common slips

- Guessing from fully simplified fractions ($\frac{3}{4}, \frac{2}{3}, \frac{5}{8}, \frac{3}{5}$) and missing the pattern: uncancelling to a common form is the trick.
- In (ii), inducting on n instead of k (possible, but messier), or forgetting that the step from k to $k+1$ only needs to run while $k+1 \leq n-1$.

Beyond STEP

Part (i) is the start of the infinite product $\prod_{m\geq 2}\left(1-\frac{1}{m^2}\right) = \frac{1}{2}$. It is a special case of Euler's product $\frac{\sin \pi x}{\pi x} = \prod_{m\geq 1}\left(1-\frac{x^2}{m^2}\right)$: divide both sides by the $m=1$ factor $1-x^2$ and let $x \to 1$. Making sense of infinite products and when they converge is a topic in first-year analysis. Part (ii) is one of many binomial identities that follow from Pascal's rule or from generating functions: it is the coefficient of x^k in $(1-x)^n \cdot \frac{1}{1-x} = (1-x)^{n-1}$. Abbott's *Understanding Analysis* is a good place to read about convergence of infinite sums and products.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.